

TrES-5b

R-0974

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_230912 · created 2026-07-29T18:07:44+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460199.7887 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.054
Transit depth (Rp/Rs)^2	3.25 +/- 1.93 [%]
Orbital Inclination (inc)	89.55 +/- 2.85
Airmass coefficient 1 (a1)	2.14 +/- 1.05
Airmass coefficient 2 (a2)	-0.7 +/- 1.28
Scatter in the residuals of the lightcurve fit is	42.69 %
Best Comparison Star	None
Optimal Aperture	0.99
Optimal Annulus	5.75
Transit Duration (day)	0.0501 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.054 vs 0.142 (deviation 0.57σ)
Duration fit vs published	0.0501 d vs 0.0758 d (deviation 3.93σ)
Mid-time O–C	8.8 min
Depth SNR	2.7
Residual scatter	42.69 %

Light curve

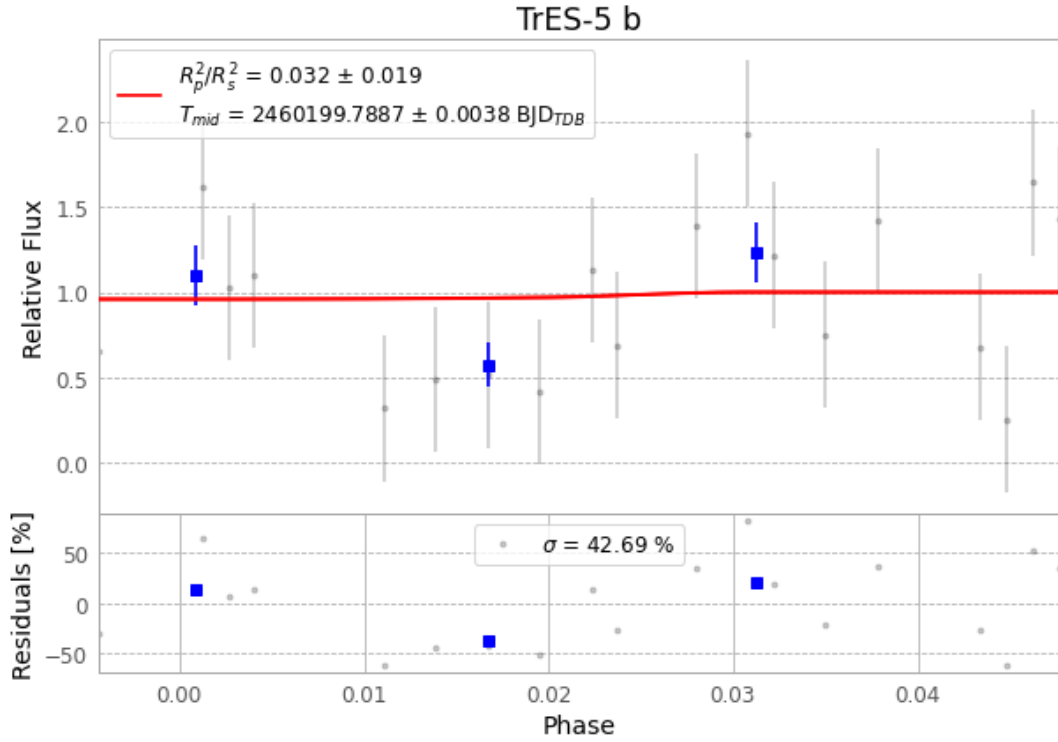


Figure 1 — FinalLightCurve_TrES-5 b_2023-09-11.png (SHA-256 a6296c45c2394c4a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 369], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a38b86e2afa7e976...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

38 FITS frames (25 MB), TRES-5230912064215.FITS → TRES-5230912083315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-230912.log (f39d1e4b0050...), FinalLightCurve_TrES-5 b_2023-09-11.png (a6296c45c239...), FinalLightCurve_TrES-5 b_2023-09-11.csv (c7cc6a1ec324...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 18:07Z.